

S&T2025

Bridging Physics

Lecture 0

Course Orientation

A/Prof Tay Seng Chuan

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The module is designed to provide a clear and logical introduction to the concepts and principles of mechanics, thermodynamics, electricity and magnetism with illustrations based on applications to the real world. Topics for mechanics include motion in one dimension, curvilinear motion, circular motion, relative motion, Newton's laws; friction, work and energy, conservative forces, conservation of energy, linear momentum and conservation, collisions, simple harmonic motion, rotational kinematics, moment of inertia and torque, rotational dynamics, conservation of angular momentum, and topics for heat and thermodynamic include temperature and the zeroth law, temperature scales, thermal expansion of solids and liquids, heat and internal energy, specific heat capacities, latent heat, work for ideal gases, first and second laws of thermodynamics, entropy and the second law, heat engines, entropy changes for reversible and irreversible processes;

Classical Mechanics

Thermodynamics

and topics included in electricity & magnetism are electric force, field & potential, current & resistance, DC circuits, electromagnetism and electromagnetic induction. These topics will sufficiently prepare you for your studies in NUS for Science and Technology courses.

Electricity and magnetism (if time permits)

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The use of Calculus

As we have studied the bridging mathematics, calculus will be used in my teaching sessions.

The details in algebraic manipulation will be skipped.

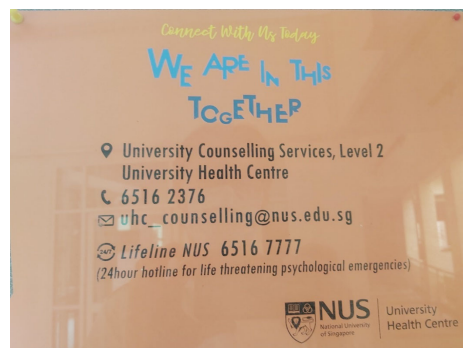
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Helplines to de-stress

We are not superman/superlady. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019

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Non-Disclosure Policy

Although the default opt-in policy will apply to scholars for counseling services, confidentiality will be strictly maintained. To preserve trust between S&T scholars and the university, the university will not disclose any information regarding the bridging course to parents if the scholars have explicitly requested that such information remain confidential, except in cases where academic warning letters must be issued.

An exception to this non-disclosure policy will apply if scholars are deemed to be at high risk of harming themselves or others, or if they exhibit suicidal tendencies.

For the mental well-being of the scholars, S&T scholars may consult a counselor [without having to inform the Warden](#). Counselors are required to adhere to the same non-disclosure policy, with exceptions only in situations where the scholar is at high risk of self-harm, harm to others, or suicide.

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Lecture 1 – Position, velocity, and acceleration of a particle in 1D

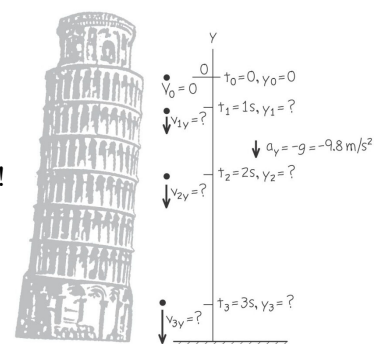
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In this lecture we will consider the motion of a particle in one dimension. A few slides in this chapter come from Chapter-2 of Young's textbook.

I conveniently present the formulations for the x -axis, but this can also be directly used for the y -axis or z -axis, **and any slanted axis!!**

As we have already studied the bridging mathematics, calculus approach will be increasingly used in the teaching. I also include additional contents not from the textbooks but they are applicable in the science and engineering domain.



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Position, velocity, and acceleration of a particle

Consider a particle of mass m , moving along the x -axis:



The **position** x of the particle changes with **time** t .

We say x is a **function** of time t , i.e., $x = x(t)$.

The **position** x has a reference called **origin** and is indicated as **O**.

The **velocity** of the particle in calculus form is $v_x = v_x(t) = \frac{d}{dt}(x(t))$.

The **acceleration** of the particle in calculus form is

$$a_x = a_x(t) = \frac{d}{dt}(v_x(t)) = \frac{d}{dt}\left(\frac{d}{dt}x(t)\right) = \frac{d^2}{dt^2}(x(t)).$$

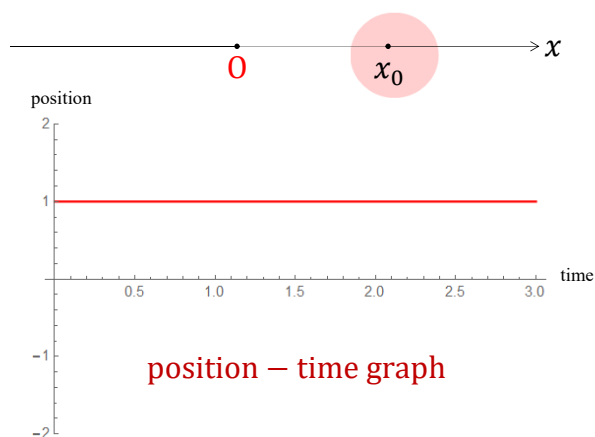
The above functions can also be expressed along the y -axis or z -axis.

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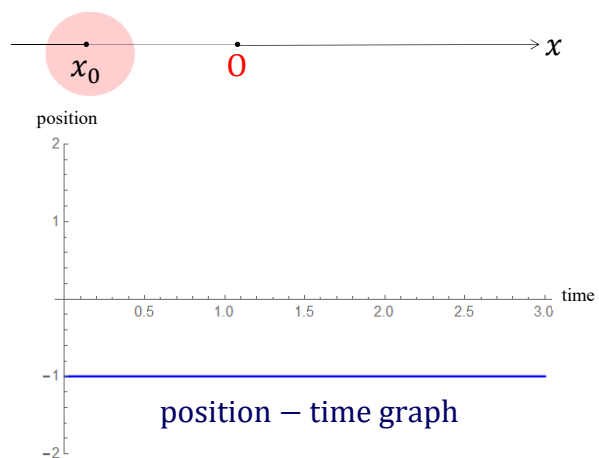
Example 1. A particle at rest at position x_0 .

$x(t) = x_0$, where x_0 is a constant and t is a variable.

If $x_0 = 1$ m, the particle will be on the **right** hand side of **O** when $t = 0$.



If $x_0 = -1$ m, the particle will be on the **left** hand side of **O** when $t = 0$.

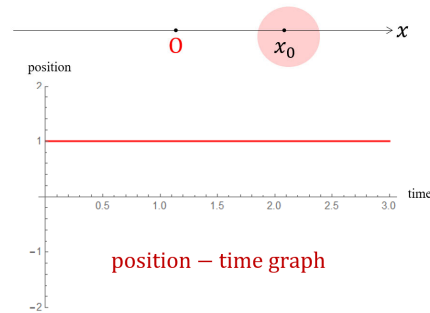


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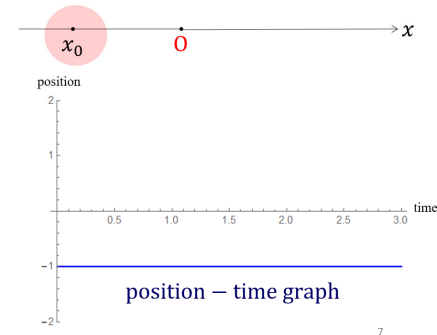
Example 1. A particle at rest at position x_0 .

$$x(t) = x_0, \text{ where } x_0 \text{ is a constant and } t \text{ is a variable.}$$

If $x_0 = 1$ m, the particle will be on the **right** hand side of **0** when $t = 0$.



If $x_0 = -1$ m, the particle will be on the **left** hand side of **0** when $t = 0$.



What if we treat that on the left of **0** is be positive?

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Example 1. A particle at rest at position x_0 (cont.)

$$x(t) = x_0, \text{ where } x_0 \text{ is a constant and } t \text{ is a variable.}$$

The **velocity** of the particle is represented as

$$v_x = v_x(t) = \frac{d}{dt}(x(t)) = \frac{d}{dt}(x_0) = \mathbf{0}.$$

The **acceleration** of the particle is represented as

$$a_x = a_x(t) = \frac{d}{dt}(v_x(t)) = \frac{d}{dt}(0) = \mathbf{0}.$$

Why **0** s?

Answer:

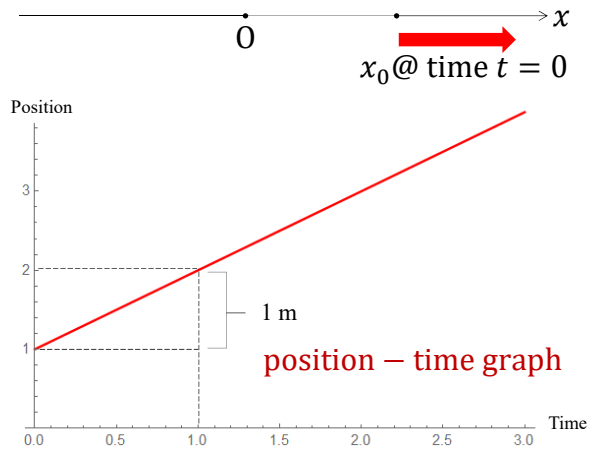
Because the particle in this function never move and never accelerate!!

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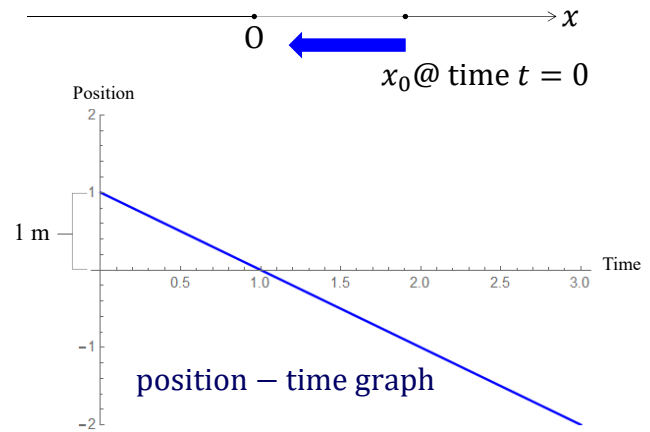
Example 2. A particle moving with constant velocity v_{0x}

$$x(t) = x_0 + v_{0x}t, \text{ where } x_0 \text{ and } v_{0x} \text{ are some constants and } t \text{ is a variable.}$$

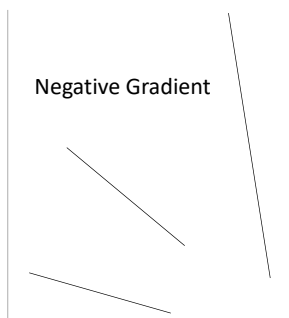
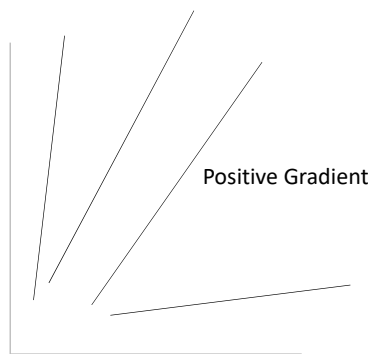
If $x_0 = 1 \text{ m}$ and $v_{0x} = 1 \text{ m/s}$,



If $x_0 = 1 \text{ m}$ and $v_{0x} = -1 \text{ m/s}$,



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Example 2. A particle moving with **constant velocity** v_{0x} (cont.)

$x(t) = x_0 + v_{0x}t$, where x_0 and v_{0x} are some constants and t is a variable.

The **velocity** of the particle,

$$v_x = v_x(t) = \frac{d}{dt}(x(t)) = \frac{d}{dt}(x_0 + v_{0x}t) = v_{0x}.$$

The **acceleration** of the particle,

$$a_x = a_x(t) = \frac{d}{dt}(v_x(t)) = \frac{d}{dt}(v_{0x}) = \mathbf{0}.$$

$$\frac{d}{dt}(t^n) = nt^{n-1}$$

Why 0?

Answer:

Although the particle in this function is moving, it never accelerate!!

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Displacement from a constant acceleration.

Since we have defined:

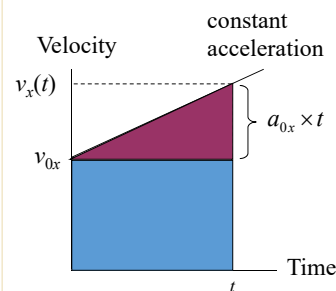
$$v_x(t) = \frac{d}{dt}(x(t))$$

$$\rightarrow \int_{t=t_1}^{t=t_2} v_x(t) dt = \int_{t=t_1}^{t=t_2} \frac{d}{dt}(x(t)) dt = x(t_2) - x(t_1) = \text{displacement}$$

Area under the Velocity versus Time graph = displacement

Area of trapezium = Area of rectangle + area of triangle

$$= t \times v_{0x} + \frac{t \times a_{0x}t}{2} = v_{0x}t + \frac{1}{2}a_{0x}t^2 \rightarrow x(t) = x_0 + \left(v_{0x}t + \frac{1}{2}a_{0x}t^2 \right)$$



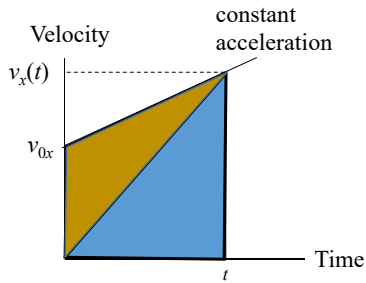
v_{0x} represents $v_x(t=0)$

a_{0x} represents the acceleration along the x -axis when time is 0, but it is a constant at any time. This will be the gradient of the slanted line.

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Displacement from a constant acceleration

As the area of the trapezium is composed by the area of 2 triangles,

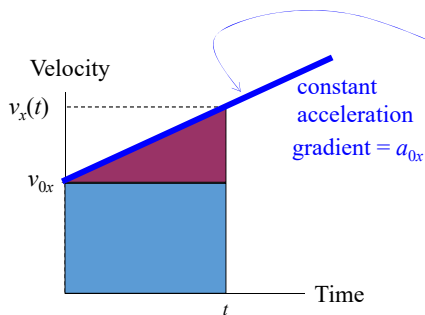


$$x(t) = \frac{v_{0x} \times t}{2} + \frac{v_x(t) \times t}{2}$$

$$= \frac{(v_{0x} + v_x(t))}{2} \times t$$

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Velocity from a constant acceleration



v_{0x} represents $v_x(t=0)$

a_{0x} represents the acceleration along the x -axis when time is 0, but it is a constant at any time. This will be the gradient of the slanted line.

From the linear equation for a straight line $y = c + mx$:

$$v_x(t) = v_{0x} + a_{0x}t$$

where

$v_x(t)$ is the final velocity at time t

v_{0x} is the initial velocity when $t = 0$

a_{0x} is the constant acceleration along a straight line (gradient of the slanted line)

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Example 3. A particle moving with constant acceleration a_{0x}

$$x(t) = x_0 + \left(v_{0x}t + \frac{1}{2}a_{0x}t^2 \right),$$

where x_0 , v_{0x} and a_{0x} are some constants.

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

x_0 is the **position** of the particle at time $t = 0$, i. e., $x(0) = x_0$.

The **velocity** of the particle, $v_x = v_x(t) = \frac{d}{dt}(x(t))$

$$= \frac{d}{dt} \left(x_0 + v_{0x}t + \frac{1}{2}a_{0x}t^2 \right) = 0 + v_{0x} + \frac{1}{2}(2a_{0x}t) = v_{0x} + a_{0x}t.$$

Consistent result!!

v_{0x} is the **velocity** of the particle at time $t = 0$, i. e., $v_x(0) = v_{0x}$.

The **acceleration** of the particle,

$$a_x = a_x(t) = \frac{d}{dt}(v_x(t)) = \frac{d}{dt}(v_{0x} + a_{0x}t) = a_{0x}.$$

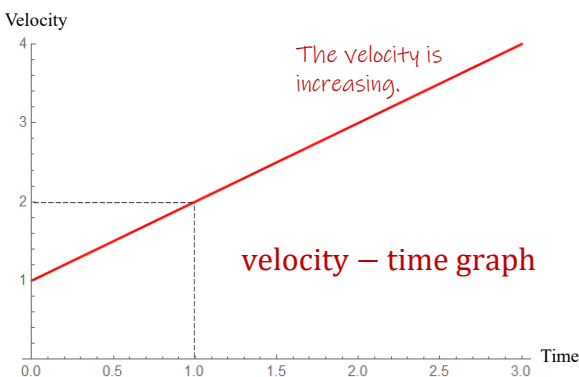
Consistent result!!

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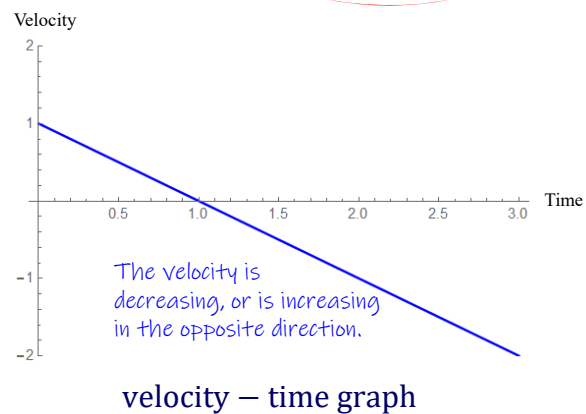
Example 3. A particle moving with constant acceleration a_{0x} (cont.)

$$v_x = v_x(t) = v_{0x} + a_{0x}t$$

If $v_{0x} = 1 \text{ m/s}$ while $a_{0x} = 1 \text{ m/s}^2$,



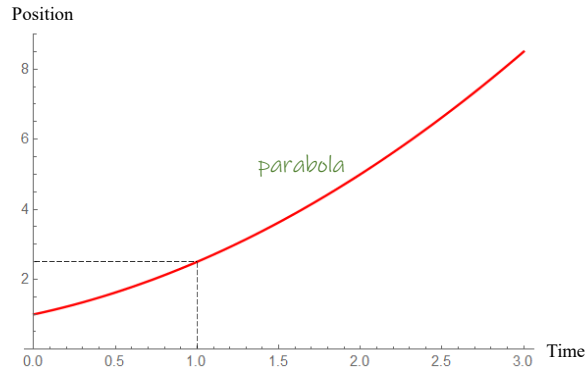
If $v_{0x} = 1 \text{ m/s}$ while $a_{0x} = -1 \text{ m/s}^2$,



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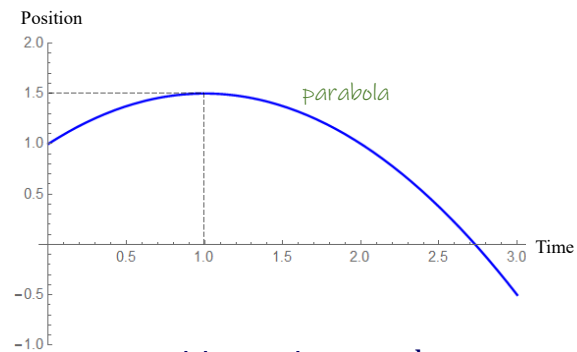
Example 3. A particle moving with constant acceleration a_{0x} (cont.)

$$x(t) = x_0 + v_{0x}t + \frac{1}{2}a_{0x}t^2$$



position – time graph

Here, $x_0 = 1$ m, $v_{0x} = 1$ m/s,
and $a_{0x} = 1$ m/s².



position – time graph

Here, $x_0 = 1$ m, $v_{0x} = 1$ m/s,
and $a_{0x} = -1$ m/s².

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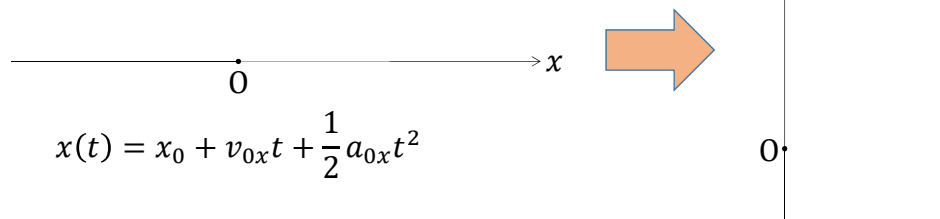
Example 3. A particle moving with constant acceleration a_{0x} (cont.)

Remark 1.

A **freely falling particle** has **position**,

$$y(t) = y_0 + v_{0y}t + \frac{1}{2}a_{0y}t^2, \text{ with } a_{0y} = -g.$$

Here, **acceleration due to gravity (near the earth surface)** $g = 9.80$ m/s².



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Why didn't the ball go to the Moon?



$$v_y(t) = v_{0y} + a_{0y}t = v_{0y} + (-9.8)t$$

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Example 3. A particle moving with constant acceleration a_{0x} (cont.)

Remark 2.

$$x = x(t) = x_0 + v_{0x}t + \frac{1}{2}a_{0x}t^2$$

$$v_x = v_x(t) = v_{0x} + a_{0x}t$$

$$\rightarrow t = \frac{v_x - v_{0x}}{a_{0x}}$$

Substituting $t = \frac{v_x - v_{0x}}{a_{0x}}$ into $x(t)$, we have

$$x = x_0 + v_{0x}\left(\frac{v_x - v_{0x}}{a_{0x}}\right) + \frac{1}{2}a_{0x}\left(\frac{v_x - v_{0x}}{a_{0x}}\right)^2 \rightarrow x - x_0 = v_{0x}\left(\frac{v_x - v_{0x}}{a_{0x}}\right) + \frac{1}{2}a_{0x}\left(\frac{v_x - v_{0x}}{a_{0x}}\right)^2$$

Multiplying through with $2a_{0x}$, we have

$$\begin{aligned} 2a_{0x}(x - x_0) &= 2v_{0x}(v_x - v_{0x}) + (v_x - v_{0x})^2 \\ &= [2v_{0x} + (v_x - v_{0x})](v_x - v_{0x}) \\ &= (v_x + v_{0x})(v_x - v_{0x}) = v_x^2 - v_{0x}^2 \end{aligned}$$

$$\rightarrow v_x^2 = v_{0x}^2 + 2a_{0x}(x - x_0)$$

Last formula for translational kinematics!!



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Formulae for Translational Kinematics with Constant Acceleration

$$v_x(t) = v_{0x} + a_{0x}t$$

$$x(t) = x_0 + \left(v_{0x}t + \frac{1}{2}a_{0x}t^2 \right)$$

$$x(t) = x_0 + \frac{(v_{0x} + v_x(t))}{2} \times t$$

$$v_x^2 = v_{0x}^2 + 2a_{0x}(x - x_0)$$

Formal version with specified axis

$$v_f = v_i + at$$

$$\Delta x = v_i t + \frac{1}{2}at^2$$

$$\Delta x = \frac{(v_i + v_f)}{2} \times t$$

$$v_f^2 = v_i^2 + 2a\Delta x$$

f : final
 i : initial
 Δx : displacement

Simplified version

$$v = u + at$$

$$s = s_0 + \frac{1}{2}(u + v)t$$

$$s = s_0 + ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2a(s - s_0)$$

v : final velocity

u : initial velocity

a : acceleration

t : time

s : final position

s_0 : initial position

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Finding the roots of a quadratic equation

<https://www.mathsisfun.com/algebra/quadratic-equation.html>

Hidden Quadratic Equations!

As we saw before, the **Standard Form** of a Quadratic Equation is

$$ax^2 + bx + c = 0$$

But sometimes a quadratic equation doesn't look like that!

For example:

In disguise		In Standard Form	a, b and c
$x^2 = 3x - 1$	Move all terms to left hand side	$x^2 - 3x + 1 = 0$	$a=1, b=-3, c=1$
$2(w^2 - 2w) = 5$	Expand (undo the brackets), and move 5 to left	$2w^2 - 4w - 5 = 0$	$a=2, b=-4, c=-5$
$z(z-1) = 3$	Expand, and move 3 to left	$z^2 - z - 3 = 0$	$a=1, b=-1, c=-3$

How To Solve Them?

Or we can use the special **Quadratic Formula**:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Just plug in the values of a , b and c , and do the calculations.

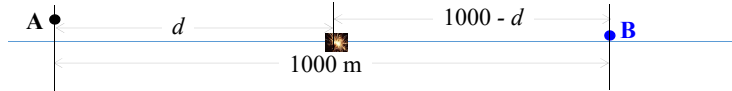
We will look at this method in more detail now.

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Question. Two cars start from points **A** and **B** which are 1000 m apart on a straight horizontal road. The first car starts from rest from **A** and moves towards **B** with a constant acceleration of 3 m/s^2 . The second car starts from **B** with a velocity of 10 m/s towards **A** and accelerates in the same direction at 1.5 m/s^2 constantly. Find the time when the cars pass one another.

Answer:

Let d be the distance travelled by the car from point **A** before they pass one another.



Use $x(t) = x_0 + v_{0x}t + \frac{1}{2}a_{0x}t^2$

From **A**: $d = 0 + 0 \times t + \frac{1}{2}(3)t^2 \rightarrow d = 1.5t^2 \dots (1)$

From **B**: $(1000 - d) = 0 + 10t + \frac{1}{2}(1.5)t^2 \rightarrow d = -0.75t^2 - 10t + 1000 \dots (2)$

$(1) = (2)$: $1.5t^2 = -0.75t^2 - 10t + 1000 \rightarrow 2.25t^2 + 10t - 1000 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$t = \frac{-10 \pm \sqrt{10^2 - 4(2.25)(1000)}}{2 \times 2.25} = \frac{-10 \pm \sqrt{9100}}{4.5} = \frac{-10 \pm 95.3939}{4.5} = \boxed{18.98 \text{ sec}}$

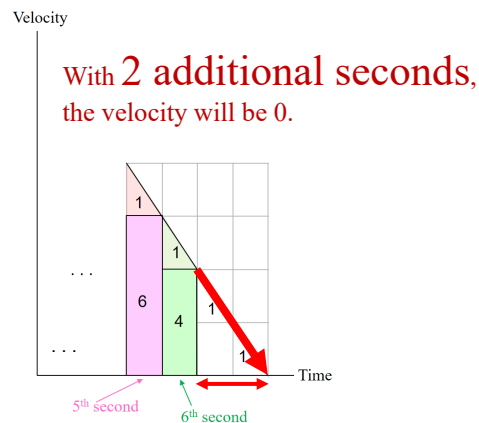
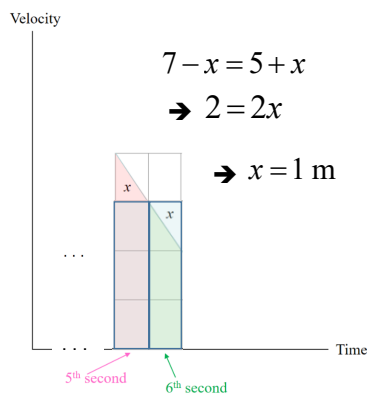
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But sometimes a simple reasoning will do

Question. A particle moving along a straight line under uniform deceleration is displaced 7 m in the 5th second, and is displaced 5 m in the 6th second. Calculate the additional time required for the particle to come to rest.

Answer:

From the following Velocity versus Time diagram with the 2 conditions given in the question:



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What if the acceleration is a variable?

Example 4. Sinusoidal functions of time. $x(t) = r_0 \cos(\theta_0 + \omega_0 t)$, where r_0, θ_0 and ω_0 are some constants.

At time $t = 0$, the **position** of the particle $x(0) = r_0 \cos \theta_0$.

The **velocity** of the particle, $v_x = v_x(t) = \frac{d}{dt} x(t)$

$$\frac{d}{dt} x(t) = \frac{d}{dt} [r_0 \cos(\theta_0 + \omega_0 t)] = (-r_0 \sin(\theta_0 + \omega_0 t)) \omega_0 = -r_0 \omega_0 \sin(\theta_0 + \omega_0 t).$$

chain rule

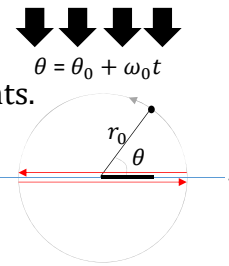
At time $t = 0$, the velocity of the particle $v_x(0) = -r_0 \omega_0 \sin \theta_0$.

The **acceleration** of the particle, $a_x = a_x(t) = \frac{d}{dt} (v_x(t)) = \frac{d}{dt} [-r_0 \omega_0 \sin(\theta_0 + \omega_0 t)]$

$$= (-r_0 \omega_0 \cos(\theta_0 + \omega_0 t)) \omega_0 = (-\omega_0^2 (r_0 \cos(\theta_0 + \omega_0 t))) = -\omega_0^2 x(t).$$

chain rule

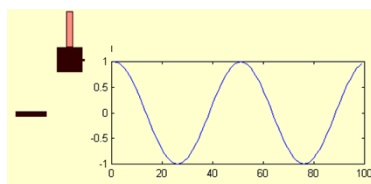
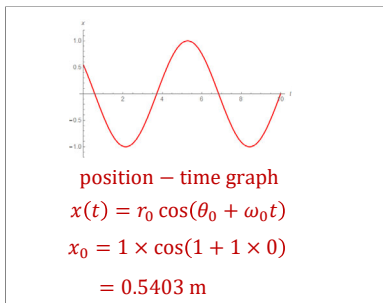
Simple Harmonic Motion



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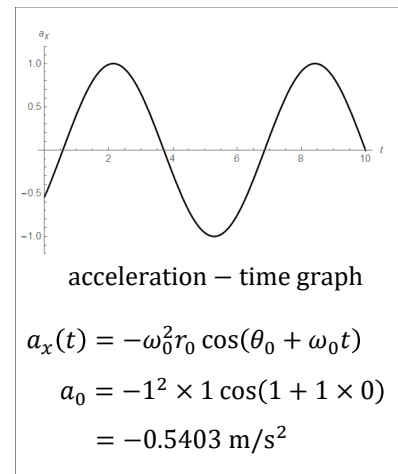
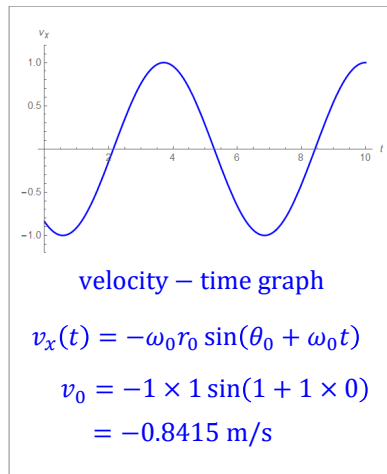
Example 4. Sinusoidal functions of time. (cont.)

If $r_0 = 1 \text{ m}$, $\theta_0 = 1 \text{ rad}$, and $\omega_0 = 1 \text{ rad/s}$, then

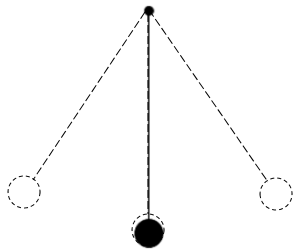


Paper

Simple Harmonic Motion



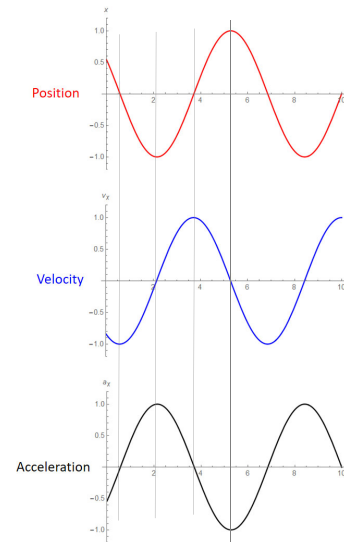
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Simple Harmonic Motion (SHM)

Other Options:

- Damped oscillation
- Forced oscillation (Resonance)



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Example 5. Exponential function of time.

$$y(t) = y_0 - \frac{g}{\gamma}t + \frac{v_{0y}}{\gamma}[1 - e^{-\gamma t}] + \frac{g}{\gamma^2}[1 - e^{-\gamma t}]$$

where $g = 9.80 \text{ m/s}^2$ and $\gamma > 0$ is some constant, and at time $t = 0$ the **position** of the particle $y(0) = y_0$.

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(e^{f(x)}) = e^{f(x)} \times \frac{d}{dx}(f(x))$$

The **velocity** of the particle,

$$\begin{aligned} v_y = v_y(t) &= \frac{d}{dt}(y(t)) = -\frac{g}{\gamma} + \frac{v_{0y}}{\gamma}(-\gamma(-e^{-\gamma t})) + \frac{g}{\gamma^2}(-\gamma(-e^{-\gamma t})) \\ &= -\frac{g}{\gamma} + v_{0y}e^{-\gamma t} + \frac{g}{\gamma}e^{-\gamma t} \end{aligned}$$

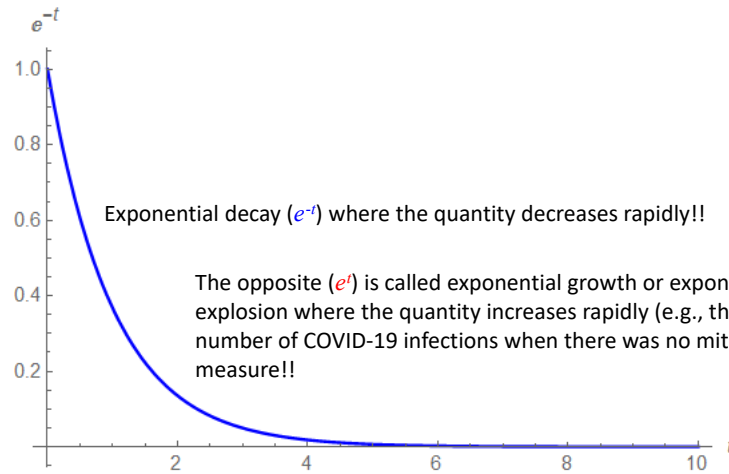
At time $t = 0$, the **velocity** of the particle is $v_y(0) = -\frac{g}{\gamma} + v_{0y} + \frac{g}{\gamma} = v_{0y}$

As time $t \rightarrow \infty$, the **velocity** of the particle is called the **terminal velocity**.

$$v_y(t \rightarrow \infty) = -\frac{g}{\gamma} + v_{0y}e^{-\gamma \times \infty \rightarrow 0} + \frac{g}{\gamma}e^{-\gamma \times \infty \rightarrow 0} = -\frac{g}{\gamma}$$

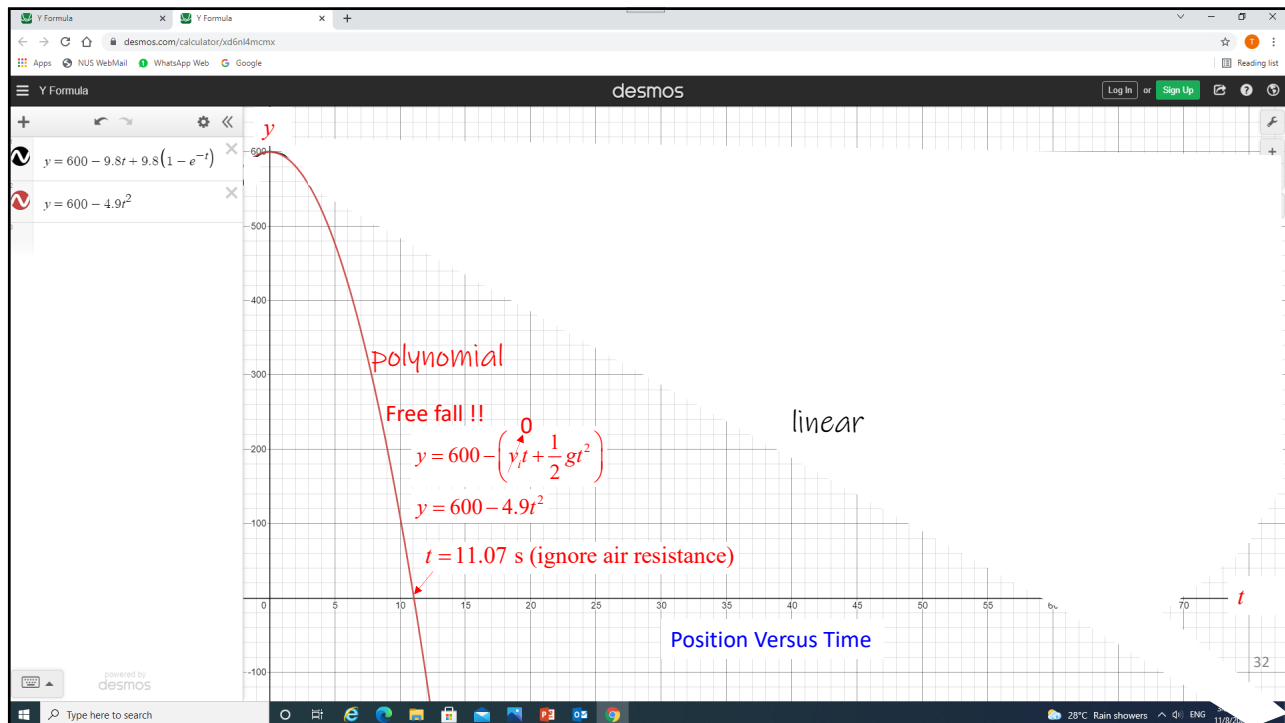
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Example 5. Exponential function of time. (cont.)

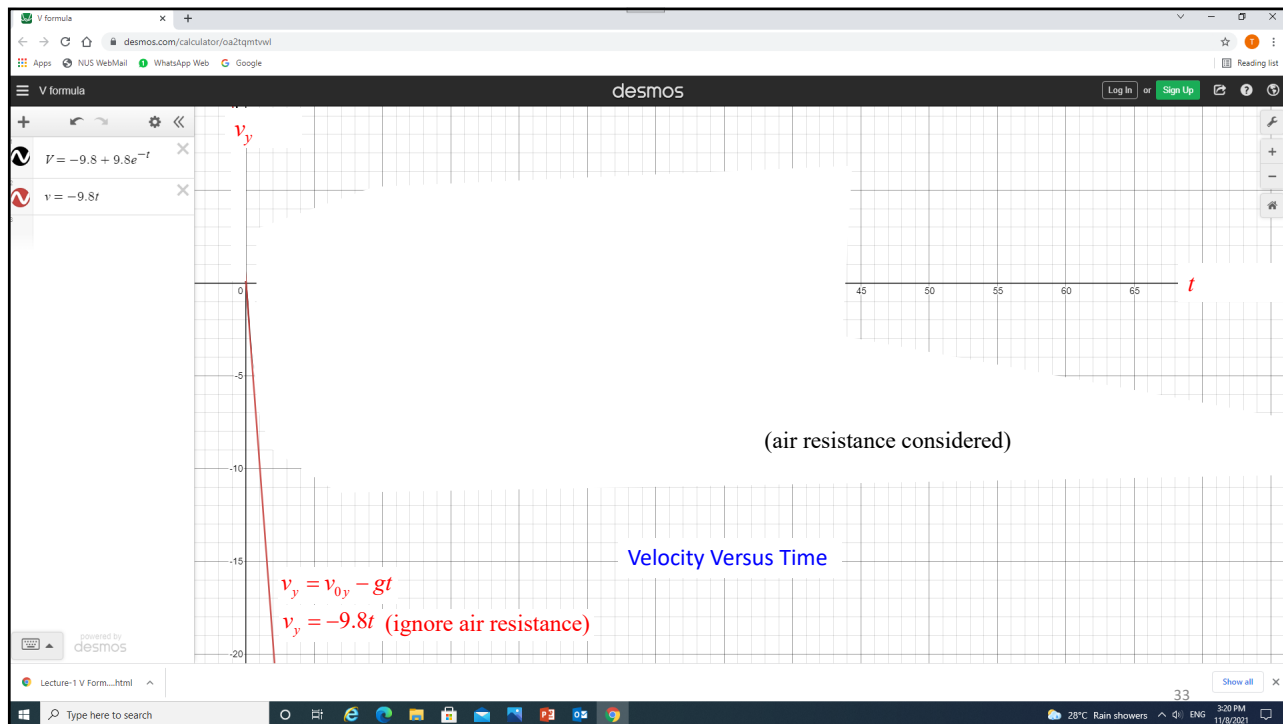


Here, $\gamma = 1$ and the **exponential function** $e^{-\gamma t} = e^{-t}$.

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Example 5. Exponential function of time. (cont.)

From $v_y = \frac{d}{dt}(y(t)) = -\frac{g}{\gamma} + v_{0y}e^{-\gamma t} + \frac{g}{\gamma}e^{-\gamma t}$

the **acceleration** of the particle, $a_y = a_y(t) = \frac{d}{dt}(v_y(t)) = v_{0y}(-\gamma e^{-\gamma t}) + \frac{g}{\gamma}(-\gamma e^{-\gamma t})$
 $= -\gamma v_{0y}e^{-\gamma t} - ge^{-\gamma t} \dots (1)$

At time $t = 0$, the **acceleration** of the particle

$$a_y(0) = -\gamma v_{0y}e^{-\gamma \times 0} - ge^{-\gamma \times 0} = -\gamma v_{0y} - g.$$

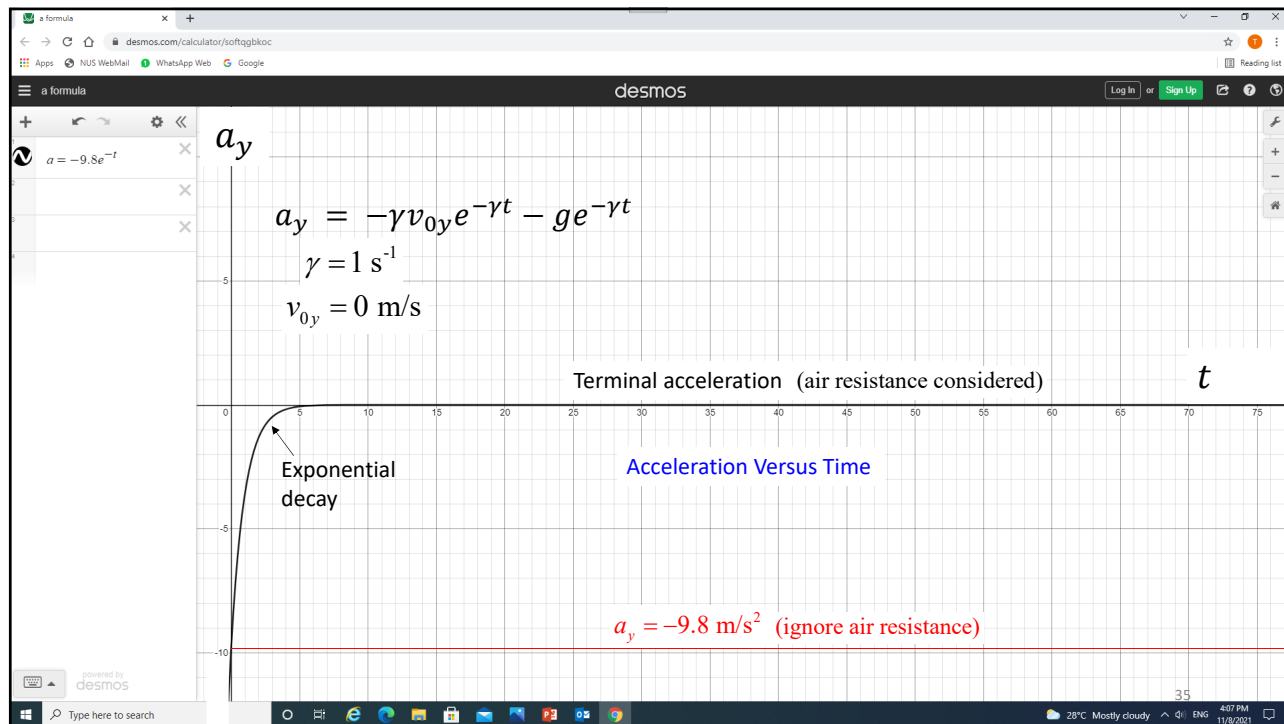
As time $t \rightarrow \infty$, the **acceleration** of the particle

$$a_y(t \rightarrow \infty) = -\gamma v_{0y}e^{-\gamma \times \infty} - ge^{-\gamma \times \infty} \rightarrow 0.$$

From $v_y \times \gamma \Rightarrow \gamma v_y(t) = -g + \gamma v_{0y}e^{-\gamma t} + ge^{-\gamma t} \dots (2)$

$$(1) + (2) \Rightarrow a_y(t) + \gamma v_y(t) = -g \Rightarrow a_y(t) = -\gamma v_y(t) - g.$$

When $a_y(t) \rightarrow 0$, $v_y(t) \rightarrow -\frac{g}{\gamma}$. **terminal velocity**



Summary of Calculus: Acceleration, velocity, and position of a particle

The **acceleration** of the particle, $a_x = a_x(t) = \frac{d}{dt}(v_x(t))$.

So, the **velocity** of the particle at time t is given by $v_x(t) = v_{0x} + \int_0^t a_x(t') dt'$.

The **velocity** of the particle, $v_x = v_x(t) = \frac{d}{dt}(x(t))$.

So, the **position** of the particle at time t is given by $x(t) = x_0 + \int_0^t v_x(t') dt'$.

In a continuous small change, final value = initial value + small offset!!

There is no why as we have simply accepted it.

Summary:

A particle moving with **constant acceleration** a_{0x}

$a_x(t) = a_{0x}$, where a_{0x} is a constant.

At time $t = 0$, if the **velocity** of the particle is $v_x(0) = v_{0x}$, some constant, then

$$\begin{aligned} v_x(t) &= v_{0x} + \int_0^t a_x(t') dt' = v_{0x} + \int_0^t a_{0x} dt' \\ &= v_{0x} + [a_{0x} t']_0^t = v_{0x} + a_{0x} t. \end{aligned}$$

Same result !!

At time $t = 0$, if the **position** of the particle is $x(0) = x_0$, some constant, then

$$\begin{aligned} x(t) &= x_0 + \int_0^t v_x(t') dt' = x_0 + \int_0^t (v_{0x} + a_{0x} t') dt' \\ &= x_0 + \left[v_{0x} t' + \frac{1}{2} a_{0x} t'^2 \right]_0^t = x_0 + v_{0x} t + \frac{1}{2} a_{0x} t^2. \end{aligned}$$

Same result !!

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Question. A Grand Prix car on the straightway along the x -axis at any

time t is given by the equation $v_x = 60 \text{ m/s} + (0.5 \text{ m/s}^3)t^2$.

What if the engine is too powerful?



- (i) Find the **change** in x -velocity of the car in the time interval $t_1 = 1.0 \text{ s}$ to $t_2 = 3.0 \text{ s}$.
- (ii) Find the average x -acceleration in this time interval.
- (iii) Find the instantaneous x -acceleration at $t_1 = 1.0 \text{ s}$ by taking Δt to be first 0.1 s , then 0.01 s , and then 0.001 s .
- (iv) Derive an expression for the instantaneous x -acceleration as a function of time, and use it to find a_x at $t = 1.0 \text{ s}$ and $t = 3.0 \text{ s}$.

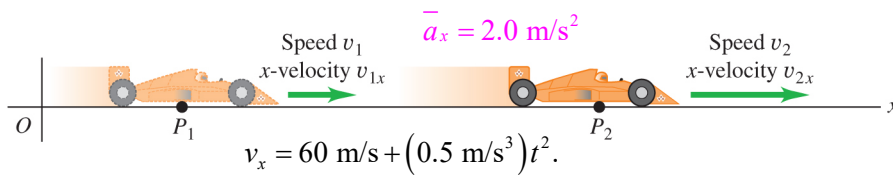
Answer:

$$\begin{aligned} \text{(i)} \quad \Delta v_x &= (60 \text{ m/s} + (0.5 \text{ m/s}^3)(3 \text{ s})^2) - (60 \text{ m/s} + (0.5 \text{ m/s}^3)(1 \text{ s})^2) \\ &= (0.5 \text{ m/s}^3)((3 \text{ s})^2 - (1 \text{ s})^2) = 4.0 \text{ m/s} \end{aligned}$$

$$\text{(ii)} \quad \bar{a}_x = \frac{\Delta v_x}{\Delta t} = \frac{4.0 \text{ m/s}}{3 \text{ s} - 1 \text{ s}} = 2.0 \text{ m/s}^2$$

As the average acceleration and the velocity have the same sign, the car speeds up.

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- (iii) Find the instantaneous x -acceleration at $t_1 = 1.0 \text{ s}$ by taking Δt to be first 0.1 s , then 0.01 s , and then 0.001 s .

- (iv) Derive an expression for the instantaneous x -acceleration as a function of time, and use it to find a_x at $t = 1.0 \text{ s}$ and $t = 3.0 \text{ s}$.

Answer:

$$(iii) \quad a_x(t=1) = \frac{\delta v_x(t=1)}{\delta t} = \frac{v_x(t=1+\delta t) - v_x(t=1)}{\delta t} = \frac{0.5(1+\delta t)^2 - 0.5 \times 1^2}{\delta t} = \frac{0.5((1+\delta t)^2 - 1)}{\delta t}$$

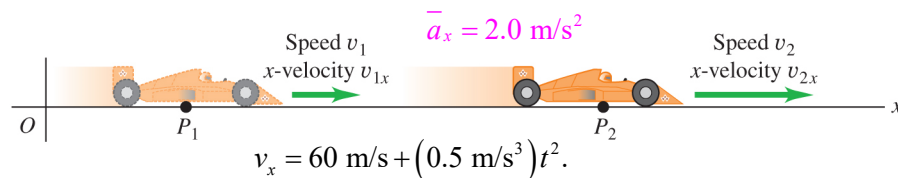
$$\text{Sub } \delta t = 0.1 \text{ s} \rightarrow a_x(t=1) = \frac{0.5((1+0.1)^2 - 1)}{0.1} = 1.05 \text{ m/s}^2$$

$$\text{Sub } \delta t = 0.01 \text{ s} \rightarrow a_x(t=1) = \frac{0.5((1+0.01)^2 - 1)}{0.01} = 1.005 \text{ m/s}^2$$

$$\text{Sub } \delta t = 0.001 \text{ s} \rightarrow a_x(t=1) = \frac{0.5((1+0.001)^2 - 1)}{0.001} = 1.0005 \text{ m/s}^2$$

From this trend
the **instantaneous**
 x -acceleration
at $t_1 = 1.0 \text{ s}$ is **1 m/s²**.

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- (iv) Derive an expression for the instantaneous x -acceleration as a function of time, and use it to find a_x at $t = 1.0 \text{ s}$ and $t = 3.0 \text{ s}$.

Answer:

$$a_x = \frac{dv_x}{dt} = \frac{d(60 \text{ m/s} + (0.5 \text{ m/s}^3)t^2)}{dt} = 2(0.5 \text{ m/s}^3)t = t \text{ m/s}^2$$

$$\rightarrow a_x(t=1) = 1 \text{ m/s}^2 \quad \text{and} \quad a_x(t=3) = 3 \text{ m/s}^2.$$

The instantaneous x -acceleration in this question is not a constant. The rate of the change of acceleration with time is sometimes called “jerk”.

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Question. Cars A and B travel in a straight line. The distance of A from the starting point is given as a function of time t by $x_A(t) = \alpha t + \beta t^2$ with $\alpha = 2.60$ m/s and $\beta = 1.2$ m/s². The distance of B from the starting point is $x_B(t) = \gamma t^2 - \delta t^3$ with $\gamma = 2.80$ m/s² and $\delta = 0.2$ m/s³.

(i) Which car is ahead immediately after the two cars leave the starting point?

Answer:

$$v_A(t) = \frac{d}{dt}(x_A(t)) = \frac{d}{dt}(\alpha t + \beta t^2) = \alpha + 2\beta t$$

$$\rightarrow v_A(t=0) = \alpha = 2.6 \text{ m/s}$$

100% Calculus Approach!!

$$v_B(t) = \frac{d}{dt}(x_B(t)) = \frac{d}{dt}(\gamma t^2 - \delta t^3) = 2\gamma t - 3\delta t^2$$

$$\rightarrow v_B(t=0) = 0$$

Thus car A is ahead of car B immediately after the two cars leave the starting point.

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(ii) At what time(s) are the cars at the same point after leaving the started point?

Answer:

$$x_A(t) = \alpha t + \beta t^2$$

$$x_B(t) = \gamma t^2 - \delta t^3$$

The cars will be at same point when their position is the same, i.e.,

$$x_A(t) = x_B(t) \quad \begin{array}{l} \alpha = 2.60 \text{ m/s and } \beta = 1.2 \text{ m/s}^2 \\ \gamma = 2.80 \text{ m/s}^2 \text{ and } \delta = 0.2 \text{ m/s}^3 \end{array}$$

$$\rightarrow \alpha t + \beta t^2 = \gamma t^2 - \delta t^3$$

$$\rightarrow 2.6t + 1.2t^2 = 2.8t^2 - 0.2t^3$$

$$\rightarrow 0.2t^3 - 1.6t^2 + 2.6t = 0$$

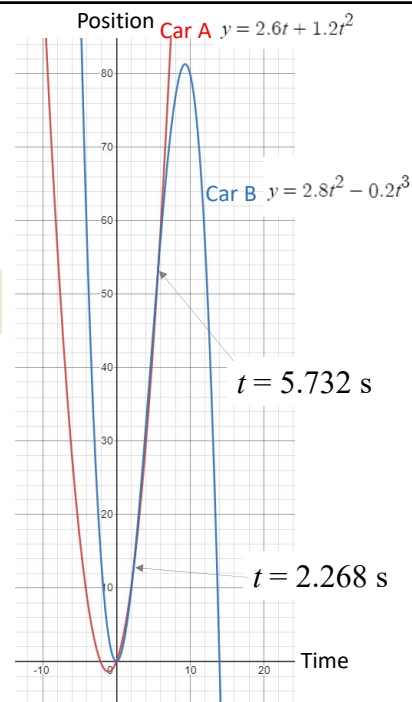
$$t(0.2t^2 - 1.6t + 2.6) = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{-(-1.6) \pm \sqrt{(-1.6)^2 - 4(0.2)(2.6)}}{2(0.2)}$$

$$= \frac{1.6 \pm \sqrt{0.48}}{0.4} = \frac{1.6 \pm 0.6928}{0.4}$$

$$\rightarrow t = 5.732 \text{ s}, 2.268 \text{ s}$$



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(iii) At what time(s) is the distance from A to B neither increasing nor decreasing?

$$v_A(t) = \alpha + 2\beta t$$

$$v_B(t) = 2\gamma t - 3\delta t^2$$

$$\alpha = 2.60 \text{ m/s and } \beta = 1.2 \text{ m/s}^2$$

$$\gamma = 2.80 \text{ m/s}^2 \text{ and } \delta = 0.2 \text{ m/s}^3$$

Answer:

Two cars will maintain the same separation **when their velocity is the same**, i.e., $v_A(t) = v_B(t)$

$$\rightarrow \alpha + 2\beta t = 2\gamma t - 3\delta t^2$$

$$\rightarrow 2.6 + 2.4t = 5.6t - 0.6t^2$$

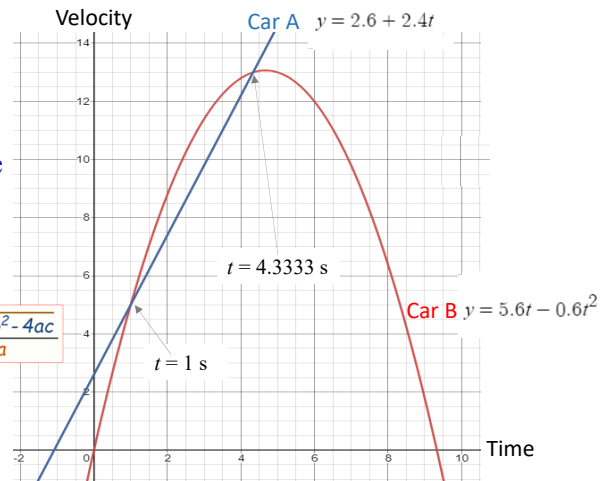
$$0.6t^2 - 3.2t + 2.6 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{-(-3.2) \pm \sqrt{(-3.2)^2 - 4(0.6)(2.6)}}{2(0.6)}$$

$$= \frac{3.2 \pm \sqrt{4}}{1.2} = \frac{3.2 \pm 2}{1.2}$$

$$\rightarrow t = 4.3333 \text{ s, } 1 \text{ s}$$



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(iv) At what time(s) do A and B have the same acceleration?

Answer:

$$v_A(t) = \alpha + 2\beta t$$

$$v_B(t) = 2\gamma t - 3\delta t^2$$

$$\alpha = 2.60 \text{ m/s and } \beta = 1.2 \text{ m/s}^2$$

$$\gamma = 2.80 \text{ m/s}^2 \text{ and } \delta = 0.2 \text{ m/s}^3$$

Acceleration of cars is given by:

$$a_A(t) = \frac{d}{dt}(v_A(t)) = \frac{d}{dt}(\alpha + 2\beta t) = 2\beta = 2 \times 1.2 = 2.4 \text{ m/s}^2$$

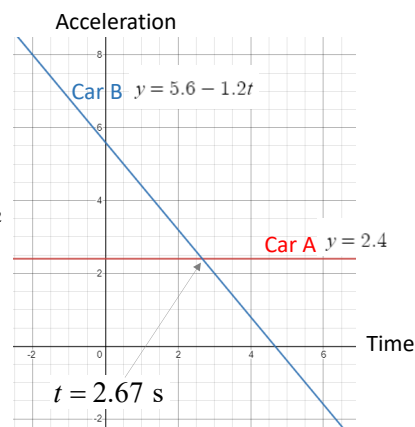
$$a_B(t) = \frac{d}{dt}(v_B(t)) = \frac{d}{dt}(2\gamma t - 3\delta t^2) = 2\gamma - 6\delta t = (5.6 - 1.2t) \text{ m/s}^2$$

At time t , both cars have same acceleration

$$\rightarrow a_A(t) = a_B(t)$$

$$2.4 = 5.6 - 1.2t$$

$$\rightarrow t = 2.67 \text{ s}$$



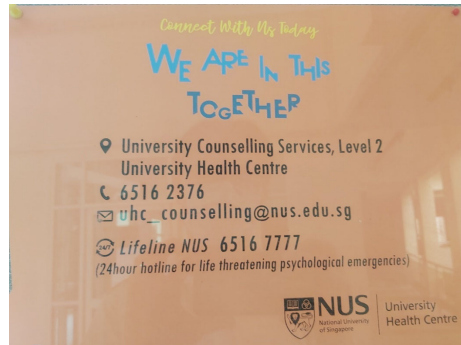
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Helplines to de-stress

We are not superman/supergirl. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019